



No notes, no computers. Language dictionaries (hard-copy) allowed.
ANSWER ALL THE QUESTIONS IN THE ANSWER BOOKLET.

1. (4 points) **Tracing**

What is the output when this code is run:

```
data = [3, -1, 7, -2, 8]
a = 0
b = 0
for c in data:
    if c < 0:
        a = c
    else:
        b = b + c
    print(f"{a}, {b}, {c}")
```

2. (4 points) **Powers of 2**

Write a Python program that prints the powers of 2 from 2^0 to 2^{100} .

```
1
2
4
....
```

3. (8 points) **Pasta**

There is unwritten rule in Italian cuisine to never break the spaghetti in half before cooking it. To have a good plate of spaghetti, **more than half** the noodles must be **15 cm long or more**.

Matteo and his friend decided to create the spaghetti-length detector. It creates a list of the length of the each individual noodle (`pasta_lengths`).

Write code that uses `pasta_lengths`

- Prints “APPROVED” if it is a good plate of pasta,
- Prints “NO!” if it is not.

Example 1:

```
pasta_lengths = [20, 28, 26, 28, 5, 29, 30] # in cm (good pasta plate)
```

Example 2:

```
pasta_lengths = [14, 18, 11, 21, 10, 5] # in cm (bad pasta plate)
```

4. (8 points) **Barnacles**

In a ocean science lab, you are tasked to count how many times the barnacles in your petri dish stick out their “tongues”. Although you start off diligently, it is a very boring task and sometimes your mind drifts. As a result, you have data that is reasonable, and data that is unreasonable.

You need to “clean” your data in the following manner:

- Your data is stored in a list called **barnacles**
 - The list contains the number of times you saw the barnacles feeding every hour.
- The average amount of feeding time has been calculated and stored in a variable **average**
- The σ (standard deviation) has also been calculated and is stored in a variable **sigma**
- In the function **clean_data**, you must **create and return** a new dataset (list) that only contains data if the value is within one standard deviation

$$|\text{value} - \text{average}| \leq \text{sigma}$$

Note: $|\text{value} - \text{average}|$ in python would be `abs(value-average)`

Example. Original data:

```
barnacles = [19, 20, 6, 21, 5, 19, 22, 20, 6, 18]
average = 16
sigma=7
```

After cleaning:

```
barnacles_clean=[19, 20, 21, 19, 22, 20, 18]
```

5. (3 points) Inputs

Here is a python program:

```
a = int(input("Give me an a: "))
b = int(input("Give me an b: "))
c = int(input("Give me an c: "))

total = 0
for i in range(5):
    total += i
    if total == a:
        print("foo")

b = b + 3
if b > 3 and b < a:
    print("bar")

if c == 2 * a + 1:
    print("garply")
```

You run this program. Give values for a, b and c that print all three messages:

```
foo
bar
garply
```

6. (8 points) **Helium Balloons**

Large helium-filled balloons are used to lift scientific equipment to high altitudes. The scientists have estimates of the pressures and temperatures at the various heights above earth (these estimates vary from day to day).

We want to calculate the volume of a balloon as it rises.

However, the balloon has a maximum size of $15m^3$. If it exceeds this limit, the balloon will burst.

- **If the volume of the balloon does not exceed $15m^3$** , then the balloon will continue to inflate according to the ideal gas law.

$\text{volume} = n * R * \text{temperature} / \text{pressure}$

where:

$n = 42$ (# of moles of air in $1m^3$ at atmospheric temperature and pressure)

$R = 8.31$

- **If the volume of the balloon exceeds $15m^3$** ,
 - print Balloon burst.
- **Print the height and volume** of the balloon for each data point unless the balloon has burst.

You do not need to format the output.

Example 2. Input data:

```
pressures = [ 100000.0, 78456.3, 60835.1, 46565.2, 35135.5, 26091.8, 19032.8 ]
heights =   [      0.0,  2000.0,  4000.0,  6000.0,  8000.0, 10000.0, 12000.0 ]
temps =     [   300.0,   287.0,   274.0,   261.0,   248.0,   235.0,   222.0 ]
```

Output:

height(m)	balloon volume (m ³)
0.0	1.0
2000.0	1.3
4000.0	1.6
6000.0	2.0
8000.0	2.5
10000.0	3.1
12000.0	4.1

Example 1. Input data:

```
pressures = [ 29357.0, 20544.6, 14008.3, 9268.2, 5919.6, 3626.0, 2111.7 ]
heights =   [ 8000.0, 10000.0, 12000.0, 14000.0, 16000.0, 18000.0, 20000.0 ]
temps =     [ 248.0, 235.0, 222.0, 209.0, 196.0, 183.0, 170.0 ]
```

Output:

height(m)	balloon volume (m ³)
8000.0	2.9
10000.0	4.0
12000.0	5.5
14000.0	7.9
16000.0	11.6
Balloon burst	

7. (6 points) **Math Functions**

Write functions for the following math expression into python. You are free to (and encouraged to) use multiple lines of code if it makes it easier to read.

$$\frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{1}{x^2 + 1}$$

$$\frac{v_i + v_f}{2} \cdot t$$

$$\sin(\omega t + p)$$

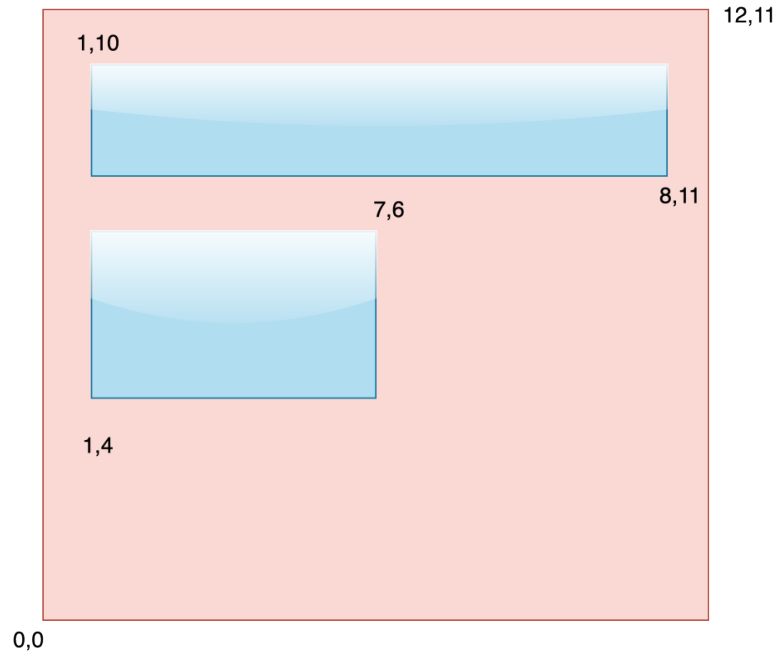
8. (8 points) **Windows**

You are wall-papering one wall in your bedroom. How much wallpaper do you need?

1. **Write a function `rect_area`** that:

- takes in the following parameters: `x1`, `y1`, `x2`, `y2`, where these are the two opposite corners of a rectangle.
- and returns the area of the rectangle.

2. **Calculate how much wall paper you need:** The wall in your room has two windows like this image:



(The coordinates for these have already been defined in the answer booklet)

In your main, use the `rect_area` function to help you calculate how many square feet of wallpaper that you need and print result.

9. (8 points) **Cars**

Jennifer would like to purchase a car and is hesitating between an electric car and a gasoline fuelled one. She would like to use a program to help her make a decision.

The program must ask input from the user for:

- The electric car's purchase price (**e_car**) in \$
- The gasoline car's purchase price(**gas_car**) in \$
- The electric car recharging cost per year (**recharge_cost**) in \$/year
- Gasoline refuelling cost per year (**refuel_cost**) in \$/year

The program must then display:

- The difference in purchase price (assume the **electric car is more expensive**)
- The difference in recharge/refuel price per year (assume the **gas car is more expensive to refuel**)
- The number of years (**int**) it would take for the savings from refuelling the electric car to be larger or equal the difference in purchase prices (assuming **refuel/recharge costs remain constant** over the years).

Example. This is only an example - your code **must be as generic as possible**. Feel free to shorten the printed messages to save time:

```
Enter the electric car purchase price: 78000
Enter the gas car purchase price: 32000
Enter the yearly recharge cost for the electric car: 500
Enter the yearly refuel cost for the has car: 2500

Purchase price difference:  46000.0
Refuel/recharge yearly savings:  2000.0
Years to make up for price diff:  23
```